

Project Proposal: Evaluating LLM Reasoning and Generation for Cancer Upstaging in Pathology Reports

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1. Relevance to NLP This project focuses on Natural Language Understanding (NLU) and complex Information Extraction (IE) within the highly specialized domain of clinical text. Analyzing unstructured pathology reports to determine cancer upstaging requires an NLP model to go beyond simple text classification. It must successfully perform multi-step logical reasoning: extracting specific entities (e.g., tumor margins, lymph node status, metastasis), understanding the temporal relationships between a baseline stage and current findings, and generating a logically sound medical deduction. This addresses a significant research question in the NLP community regarding how well large language models (LLMs) can perform reliable, multi-step logical reasoning over dense, domain-specific text without relying on black-box heuristics. By tackling this, the project produces insights into prompt engineering and model reliability that are of high interest to both the NLP and healthcare AI research communities.

2. Current State of the Research Literature and Project Contribution The current literature demonstrates that domain-specific models, such as ClinicalBERT (Alsentzer et al., 2019) and BioGPT (Luo et al., 2022), excel at zero-shot classification and named entity recognition in medical texts. Furthermore, foundational work like "Language Models are Few-Shot Learners" (Brown et al., 2020) established that generalized models can adapt to specific tasks using in-context learning. However, a major limitation of current LLMs, even those adapted to the medical domain, is their tendency to hallucinate or fail at tasks requiring comparative logic, such as determining if a cancer has "upstaged" based on new textual evidence compared to historical data.

Additionally, current research on the recently released TCGA-Path-Reports dataset (Tatonetti Lab, 2024), which provides over 9,500 machine-readable pathology reports, indicates that while LLMs can identify cancer presence, multi-variable staging remains a challenge. This project builds upon recent research into Chain-of-Thought (CoT) prompting (Wei et al., 2022). It addresses the limitation of "black-box" clinical predictions by redesigning CoT prompts to force the model to explicitly state the extracted TNM (Tumor, Node, Metastasis) values from the text before generating an upstaging conclusion. This approach aims to bridge the gap between simple entity extraction and complex clinical reasoning, thereby improving the explainability and trust of the generated output.

3. Research Questions This project aims to answer the following core research questions:

- **RQ1:** Can open-source Large Language Models (e.g., LLaMA-3) accurately extract the necessary TNM staging criteria and historical baselines from complex, unstructured pathology reports?
- **RQ2:** Does the implementation of a constrained, clinical Chain-of-Thought (CoT) prompting strategy significantly improve the model's accuracy and reliability in determining cancer upstaging compared to standard zero-shot prompting?
- **RQ3:** To what extent does enforcing an "Evidence-Grounded CoT", where the model must quote the exact span of text from the clinical note justifying its label, reduce the rate of plausible but clinically incorrect rationales (hallucinations)?

4. Experimental Design and Proposed Evaluation Metrics

Experimental Design:

- **Dataset Setup:** The project will utilize the publicly available TCGA-Path-Reports dataset, which contains 9,523 high-quality, cleaned pathology reports derived from The Cancer Genome Atlas covering various tissue types.
- **Data Preparation:** A subset of these reports will be isolated and preprocessed. Ground truth annotations for TNM staging and upstaging status will be generated. To handle the complexity of the data, preprocessing scripts will be utilized to normalize the text and prepare the splits (Train/Validation/Test) ensuring a balanced distribution of sentence lengths and cancer types across the splits.
- **Methodology:** The experiment will evaluate a quantized, open-source LLM (such as Code-LLaMA or LLaMA-3 8B, run locally to simulate secure, offline clinical deployment) under three distinct prompting conditions:
 1. *Baseline (Zero-Shot):* A standard prompt asking the model to directly classify if the pathology report indicates cancer upstaging.
 2. *Standard Few-Shot CoT:* A prompt including 3-4 examples of how to logically deduce upstaging from similar reports.
 3. *Evidence-Grounded CoT:* A novel prompting strategy requiring the model to sequentially extract the baseline stage, extract the new pathology findings, quote the supporting text span, compare the two, and then state the final conclusion.

Proposed Evaluation Metrics: The evaluation will assess both the final classification accuracy and the quality of the reasoning text generated by the model.

- **Classification Metrics:** Due to the inherent class imbalances in clinical datasets (where severe upstaging may be less frequent than stable pathology), standard accuracy is insufficient. We will utilize **Precision, Recall, and the Macro-Average F1-score** to evaluate the final upstaging determination. The Macro-Average F1 ensures both majority and minority classes are optimized equally.
- **Rationale Evaluation:** To evaluate the quality of the generated medical reasoning, we will use **BERTScore**. This metric will measure the semantic similarity between the model's generated logical steps and a human-written ground truth rationale, ensuring the meaning is clinically sound even if the exact phrasing differs.
- **Human Evaluation:** A random sample of 20-30 model outputs across the different prompting strategies will undergo a blinded human evaluation framework. The generated texts will be scored (on a scale of 1-5) based on three criteria: Clinical Safety (does the reasoning pose a risk?), Clarity, and Logical Consistency.